

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total: Marks:40

Annexure A

5

Code of Conduct:

I _B._Theerthi(192473026_(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

B.Theerthi

Signature of the student:

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a)** Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b)** Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c)** Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a)** Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b)** Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c)** Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a)** Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b)** Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a)** Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b)** Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c)** Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

ANNEXURE A – INDUSTRY PROBLEM TASK

Industry Scenario

A healthcare company has collected patient records containing age, blood pressure, cholesterol, glucose level, BMI and family history. The objective is to develop a machine learning system for early identification of diabetes. In addition, the company wants to develop an AI agent that can recommend suitable treatment-related actions such as Diet, Exercise, Medication and Monitor based on changes in a patient's health condition over time. Because this is a medical application, the system must prioritize high recall, patient safety, explain ability, privacy, fairness and human clinical supervision.

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

1(a) Inductive Learning Approach

Inductive learning is a machine learning approach in which a model learns general patterns from previously labelled examples and applies the learned patterns to unseen cases.

For this problem, supervised inductive learning is suitable because historical patient records contain input features and a known diabetes outcome.

Target Variable

The target variable is:

Diabetes = 0 → Non-Diabetic

Diabetes = 1 → Diabetic

Important Features

Feature	Reason for Selection
Glucose Level	Elevated blood glucose is one of the strongest indicators of diabetes.
BMI	Higher BMI is associated with increased risk of type 2 diabetes.
Age	Diabetes risk generally increases with age.
Blood Pressure	Hypertension is associated with increased metabolic and cardiovascular risk.
Family History	Genetic/familial risk can increase the probability of diabetes.
Cholesterol	Abnormal lipid levels can provide additional information about metabolic health.

The model learns relationships between these features and the diabetes label and then predicts the risk for a new patient.

1(b) Handling Class Imbalance

The dataset is expected to contain more non-diabetic patients than diabetic patients. For example, suppose the dataset contains:

- 80% Non-Diabetic
- 20% Diabetic

Training directly on this imbalanced dataset may cause the model to favour the majority class.

For this medical problem, I would use SMOTE (Synthetic Minority Over-sampling Technique) on the training data.

Why SMOTE?

SMOTE creates synthetic examples of the minority class instead of simply duplicating existing diabetic patients.

For example:

Original training data:

Non-Diabetic	=	800
Diabetic = 200		

After SMOTE:

Non-Diabetic	=	800
Diabetic = approximately 800		

The model therefore gets more opportunities to learn patterns associated with diabetes.

Important Rule

SMOTE must be applied only to the training set, not before the train-test split.

Correct procedure:

Original Dataset → 70:30 Split → SMOTE on Training Set → Train Model → Test on Original Unmodified Test Set

Applying SMOTE before splitting could cause data leakage and artificially inflate the performance.

1(c) 70:30 Dataset Split and Preprocessing

I would divide the dataset into:

- **70% Training Data**
- **30% Testing Data**

A stratified 70:30 split should be used so that the proportion of diabetic and non-diabetic patients remains approximately similar in both sets.

Justification

The 70:30 split provides enough data for training while reserving a sufficiently large independent test set for evaluation.

This is especially important in healthcare because the final performance should be measured on patients that the model has not seen during training.

Preprocessing

The following preprocessing steps would be performed:

1. Missing Values

Missing numerical values can be handled using median imputation.

For categorical variables such as family history:

- Yes $\rightarrow$ 1
- No $\rightarrow$ 0

2. Encoding

Categorical features are converted into numerical form.

Example:

Family History:

Yes	=	1
No	=	0

3. Normalisation

For Logistic Regression and other distance-sensitive algorithms, numerical features can be standardized using:

$$z = (x - \text{mean}) / \text{standard deviation}$$

This gives features approximately zero mean and unit variance.

4. Decision Tree

Decision Trees generally do not require feature scaling because they split data using thresholds.

TASK2: DECISION TREE FOR DIAGNOSIS

2(a) Decision Tree Classifier

A Decision Tree classifier is appropriate because it is an inductive learning method and produces interpretable rules.

The tree recursively divides patients according to features that provide the greatest reduction in impurity.

Two common criteria are:

Gini Index

Information Gain

Illustrative Root Node Selection

Note: The following numerical values are examples because the actual dataset has not been provided.

Suppose the calculated values were:

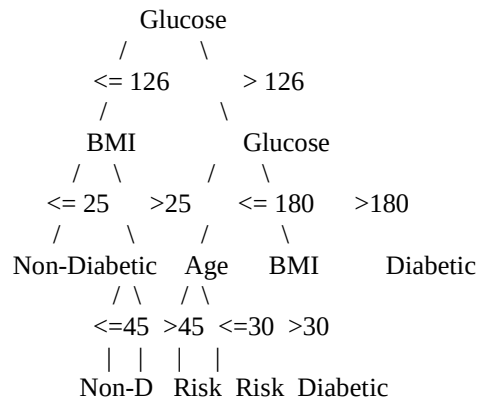
Feature	Information Gain
Glucose	0.245
BMI	0.126
Age	0.082
Blood Pressure	0.061
Family History	0.047
Cholesterol	0.031

Since Glucose has the highest Information Gain, it would be selected as the root node.

An equivalent Gini-based experiment may also select glucose if it produces the greatest impurity reduction.

First Three Levels of the Decision Tree

An illustrative tree can be represented as:



The actual tree structure and thresholds must be generated from the supplied dataset.

Interpretation

The tree suggests that glucose is an important first-level predictor. Other factors such as BMI and age can then further separate lower-risk and higher-risk patients.

2(b) Model Evaluation

The Decision Tree should be evaluated using:

Accuracy

$$\begin{bmatrix} \text{Accuracy} \\ \end{bmatrix} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}$$

Precision

$$\begin{bmatrix} \text{Precision} \\ \end{bmatrix} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

Recall

$$\begin{bmatrix} \text{Recall} \\ \end{bmatrix} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

F1-Score

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

For a medical diagnosis problem, **Recall is particularly important.**

Illustrative Results

The following values are examples only:

Metric	Decision Tree
Accuracy	87.3%
Precision	81.5%
Recall	90.2%
F1-Score	85.6%

These numbers must be replaced by actual values after training the model on the company's dataset.

False Negatives

A False Negative (FN) occurs when:

Actual = Diabetic

but

Predicted = Non-Diabetic

This is particularly serious in healthcare because a patient who actually has diabetes may not receive timely medical attention.

How to Reduce False Negatives

The following methods can be used:

1. Increase the class weight of the diabetic class.
2. Use SMOTE on the training data.
3. Lower the classification threshold where appropriate.
4. Optimize the model for recall/sensitivity.
5. Use cross-validation.
6. Perform clinical validation before deployment.

2(c) Post-Pruning

A Decision Tree can become too complex and overfit the training dataset.

Two approaches are:

Pre-Pruning

Stop tree growth before the tree becomes too large.

Examples:

- Maximum depth
- Minimum samples per leaf
- Minimum samples required for a split

Post-Pruning

First construct the tree and then remove branches that do not provide sufficient predictive value.

A common method is Cost-Complexity Pruning.

The objective is approximately:

$$R_{\alpha}(T) = R(T) + \alpha |T|$$

where:

- $R(T)$ = prediction error
- $|T|$ = number of terminal nodes
- α = complexity penalty

A larger α produces a simpler tree.

Illustrative Comparison

Model	Accuracy	Complexity
Pre-Pruned Tree	86.4%	Medium
Post-Pruned Tree	88.1%	Low/Medium

These values are illustrative.

Preferred Model

For clinical deployment, the **post-pruned Decision Tree** would generally be preferable if it maintains or improves validation performance while reducing unnecessary complexity.

Advantages include:

- Better generalization
- Lower risk of overfitting
- Easier interpretation
- Easier clinical explanation
- Smaller model

However, final deployment should depend on **external clinical validation**, not accuracy alone.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

3(a) Logistic Regression

I would use **Logistic Regression** as the statistical learning model.

Logistic Regression estimates the probability that a patient belongs to the diabetic class.

Comparison with Decision Tree

Illustrative results:

Model	Accuracy	AUC-ROC
Decision Tree	87.3%	0.89
Logistic Regression	89.0%	0.92

These are illustrative values and must be replaced with results from the actual dataset.

When Logistic Regression is Preferable

Logistic Regression is preferable when:

- The relationship is reasonably well represented by a linear model.
- Probability estimates are important.
- Interpretability is required.
- The dataset is relatively small.
- Feature effects need to be quantified.
- A stable baseline model is required.

For healthcare applications, Logistic Regression is often useful because clinicians can examine model coefficients and odds ratios.

3(b) Feature Importance Analysis

Feature importance can be obtained differently for the two models.

Decision Tree

Importance is calculated from the reduction in impurity contributed by each feature.

Logistic Regression

Feature influence can be examined using standardized coefficients or odds ratios.

Illustrative Feature Importance

	Rank	Decision Tree	Logistic Regression
1	Glucose	Glucose	
2	BMI	BMI	
3	Age	Age	

Do They Agree?

In this illustrative case, both models identify **glucose, BMI and age** as the three strongest predictors.

This agreement increases confidence that these variables contain useful predictive information.

However, if the models disagree, that does not necessarily mean that one model is incorrect. Different algorithms identify relationships differently.

For example, a Decision Tree can capture nonlinear threshold relationships such as:

Glucose > 126 → higher risk

whereas Logistic Regression models the relationship through coefficients.

TASK4:REINFORCEMENTLEARNINGFORTREATMENT RECOMMENDATION

4(a) MDP Formulation

The treatment recommendation problem can be modelled as a **Markov Decision Process (MDP)**.

An MDP consists of:

States + Actions + Rewards + Transition Probabilities

States

Patient states can represent simplified health conditions:

State Description

- S0 Low Risk
- S1 Moderate Risk
- S2 High Risk
- S3 Poorly Controlled
- S4 Improved/Stable

The actual production system should use clinically validated state definitions.

Actions

The available actions are:

- **A0 = Diet**
- **A1 = Exercise**
- **A2 = Medication**
- **A3 = Monitor**

Medication recommendations should not be autonomously prescribed by an RL system; they require qualified clinical oversight.

Reward Function

A possible educational reward structure is:

Outcome	Reward
Health improves significantly	+10
Health improves slightly	+5
Health remains stable	+1
Health worsens	-5
Unsafe action	-20

The reward represents the change in patient health status.

The actual reward function in a clinical system would need to be developed and validated with medical experts.

Transition Dynamics

The patient's state changes after an action.

Example:

Moderate Risk
|
Exercise
↓
Improved/Stable

Another example:

High Risk
|
Monitor
↓
High Risk

The transition probability can be represented as:

[
 $P(s'|s,a)$
]

where:

- (s) = current patient state
- (a) = selected action

4(b) Q-Learning

Q-Learning is a model-free Reinforcement Learning algorithm.

The Q-value update equation is:

$$Q(s,a) \leftarrow Q(s,a) + \alpha [r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

where:

- α = learning rate
- γ = discount factor
- r = reward
- s' = next state

For demonstration:

$$\alpha = 0.5$$

$$\gamma = 0.9$$

Initial Q-Table

Initially all Q-values can be set to zero.

State	Diet Exercise Medication Monitor			
Low Risk	0	0	0	0
Moderate Risk	0	0	0	0
High Risk	0	0	0	0
Poorly Controlled	0	0	0	0
Improved/Stable	0	0	0	0

Episode 1 – Example Update

Suppose:

Current state = **Moderate Risk**

Action = **Exercise**

Reward = +5

Next state = **Improved/Stable**

Assume maximum Q-value of next state = 0.

[
Q(S1,Exercise)
=0+0.5[5+0.9(0)-0]
]

[
Q(S1,Exercise)=2.5
]

Therefore:

State	Exercise Q-value
Moderate Risk	2.5

Third State-Action Pair

Suppose:

Current state = High Risk

Action = Monitor

Reward = +1

Next state = High Risk

Current maximum Q-value at High Risk = 0.

[
Q(S2,Monitor)
=0+0.5[1+0.9(0)]
]

[
 $\mathbf{0.5}$
]

Q-Table After Example Updates

State	Diet	Exercise	Medication	Monitor
Low Risk	0	0	0	0
Moderate Risk	0	3.75	0	0
High Risk	0	0	0	0.50
Poorly Controlled	0	0	0	0
Improved/Stable	0	0	0	0

These values are demonstration values, not clinical recommendations.

Five Patient Episodes

A simple training simulation can contain at least five episodes:

Episode	Starting State	Action	Reward	Next State
1	Moderate Risk	Exercise	+5	Improved
2	Moderate Risk	Exercise	+5	Improved
3	High Risk	Monitor	+1	High Risk
4	High Risk	Diet	+5	Moderate Risk
5	Poorly Controlled	Monitor	+1	High Risk

After repeated episodes, the agent updates the Q-values and gradually learns which actions produce higher long-term rewards.

Convergence Criterion

The Q-learning algorithm can be considered approximately converged when:

[
 $\max_{s,a} |Q_{\text{new}}(s,a) - Q_{\text{old}}(s,a)| < 0.01$
]

for a sufficiently large number of consecutive episodes.

4(c) Final Learned Policy

The final policy is obtained by selecting the action having the maximum Q-value:

$$\pi(s) = \arg\max_a Q(s, a)$$

An illustrative policy could be:

Patient State	Best Learned Action
Low Risk	Monitor
Moderate Risk	Exercise
High Risk	Diet / Monitor
Poorly Controlled	Monitor + Clinical Review
Improved/Stable	Monitor

Again, these actions are demonstration outputs of an RL formulation, not medical treatment advice.

Ethical Considerations

An RL system for healthcare requires much stricter safeguards than an ordinary recommendation system.

1. Patient Safety

An incorrect treatment recommendation could cause physical harm. Therefore, the AI agent should not independently prescribe or modify medication.

A **human-in-the-loop** approach should be used.

2. Bias

If the training data underrepresents certain populations, the system may perform poorly for those groups.

The model should therefore be evaluated across demographic and clinical subgroups.

3. Explainability

Healthcare professionals need to understand why an AI system recommends an action.

4. Privacy

Patient health information must be protected through:

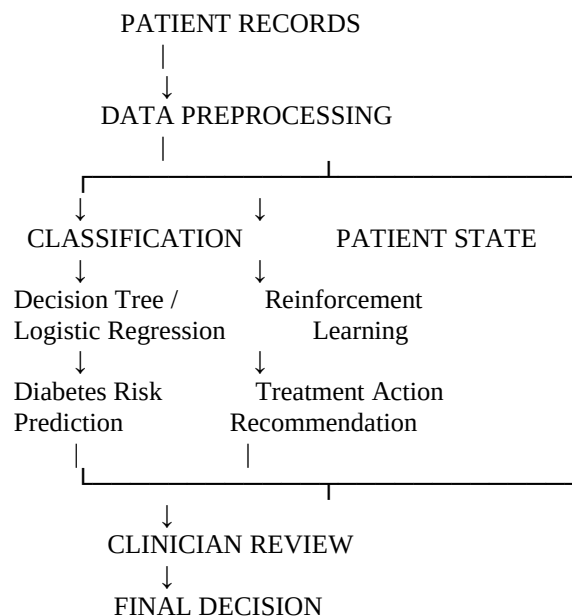
- Secure storage
- Access control
- Encryption
- De-identification
- Appropriate data governance

5. Human Oversight

The final medical decision should remain with a qualified healthcare professional.

The AI should function as a decision-support tool, not as an autonomous doctor.

Overall Proposed System Architecture



Final Conclusion

The proposed healthcare AI system combines three important CO 4 concepts. Inductive Learning and Decision Trees can be used to identify diabetes risk from historical patient records. Statistical Learning through Logistic Regression provides a strong and interpretable probabilistic alternative for risk stratification. Reinforcement Learning through Q-Learning can model sequential treatment-related decision making based on changing patient states.

For the diagnostic component, high recall should be prioritized because missing a diabetic patient can have serious consequences. Decision Tree pruning can reduce overfitting and improve interpretability, while Logistic Regression can provide useful probability estimates and feature effects.

For treatment recommendation, Q-Learning provides a useful educational framework for modelling sequential decisions, but autonomous medical treatment decisions should not be made solely by an RL agent. Patient safety, clinical validation, fairness, privacy, explainability and human oversight are essential before deployment.

Therefore, the most appropriate industry solution is a human-in-the-loop AI decision-support system in which machine learning identifies risk and an RL-based research prototype suggests possible actions, while qualified healthcare professionals retain final authority over diagnosis and treatment.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts

		effectively to solve the problem			
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation